

DIVERSIFYING INTO RENEWABLE ENERGY CHALLENGES AND OPPORTUNITIES



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INTRODUCTION

The energy transition is the move away from fossil fuels towards renewable and sustainable forms of production and generation, in combination with increasing decarbonization (net zero) and electrification. The motivations behind the energy transition are primarily political, environmental, and increasingly, financial. Mostly, it is driven by Governments and international bodies (like the EU, which also sees renewables to increase its' energy independence) through goal setting, provision of incentives, and legislation such as the US's Inflation Reduction Act.

The push for decarbonization and ESG is also now being championed by large banks and financial institutions like Barclays Bank, who recently announced that it had tightened its financing rules and abandoned financing for oil exploration altogether. Over the last 12-months or so, geopolitics has played an ever-greater role in shaping the energy industry and the energy transition, as the fall-out from the Russia-Ukraine war has interfered with the energy transition agenda, resulting in soaring power and natural gas prices. This has wrought havoc with consumers and suppliers alike and stalled, or temporarily reversed, certain net zero initiatives, and encouraged the specter of market intervention.

The energy transition impacts almost every firm in the energy sector and beyond it in a multiplicity of ways; many of whom require

additional and more detailed guidance to adequately plan without disruption to operations. In many instances, however, clarity is lacking at the detailed requirement level, and the operational realities of the energy supply and demand dynamic require even greater flexibility and agility to operate successfully. Despite that, we do know that calculating and reporting carbon footprints, and understanding the carbon intensity of various commodities, will become increasingly important, and impact corporate value in many ways. Establishing a value for carbon – its' cost – will also be an essential activity, and one that is not necessarily an easy task given there isn't a single global reference market for carbon.

All-in-all, the energy transition poses many significant challenges, as well as opportunities, to the industry. Responding to these

opportunities and challenges in a timely and adequate fashion is a key success criterion and one that requires agile systems and software solutions including CTRM (Commodity Trading and Risk Management), as well as improved

business processes. This paper reviews some of the challenges faced by the industry and looks at how CTRM (often termed ETRM in energy) and related software solutions can help in a successful outcome.

ENERGY IN TRANSITION

The energy transition is a fundamental paradigm shift requiring significant planning, consideration, and investment. It impacts not just physical players in the industry like utilities and power marketers, but also almost all entities. The transition is also comprised of any number of moving parts such that it is not only about changing from large-scale fossil fuel generation to smaller, distributed renewables. It also incorporates adding storage in form of batteries and/or hydrogen, requires investment into the grid and in smart devices, and the management of significantly increased load via electrification of households and vehicles. Furthermore, it looks for the decarbonization of supply chains, operations, and assets, along with new forms of regulatory reporting around ESG. Participation is mandatory as the drivers are largely political; none the less, this represents a huge challenge.

To add further complexity, all of this is now taking place against the backdrop of a war in the Ukraine and the many political actions taken against Russia that essentially removed a source of cheaper fossil fuels, resulting in massive price volatilities; particularly in Europe, which is already naturally short natural gas. In turn, this has facilitated political intervention in the markets in the form of price caps, government ownership, and more, as consumers watched their power prices rise exponentially. It has also meant re-examining the speed at which certain generation facilities, including coal, nuclear and natural gas, are decommissioned. The EU is now committed to finding

alternative sources to replace Russian energy, meaning that renewables have also gained in importance as one way of doing exactly that. This all adds a further layer of urgency, challenge, and opportunity to the energy transition.

The energy transition requires businesses to set strategies around renewables, manage a host of new exposures that range from market and credit risk through to regulatory and reputational risk, as well as optimize operations – not just from a cost and efficiency perspective, but also from a carbon intensity and footprint perspective. Managing smaller

scale renewable and battery assets, perhaps while also managing fossil fuel or nuclear assets that will eventually be decommissioned, is a massive challenge fraught with risk. Managing those risks, while managing operations efficiently and profitably, is the order of the day. Meanwhile, grasping the new business opportunities of being a 'green business' is also key.

The drive to net zero is a key consideration for the entire energy value chain and so creating and deploying compliance strategies is important. Depending on the nature of the firm, these could involve different issues and considerations, however, offsetting emissions with allowances is one way to go, introducing additional needs in terms of verifying, tracking, and buying the various certificates required. Recently, EU allowances exceeded 100 Eur/Tonne so managing this exposure is critical. Companies operating in the EU in many industry segments — including gas, coal power generation or industrial manufacturing — are obliged to buy such carbon credits. Each one allows the emission of 1 tonne of carbon.

The energy transition is not just about energy producers but impacts all firms in the supply chain and beyond. Some of these firms, perhaps consumers,

transporters, and so on, will find themselves affected by the need to engage in carbon and allowances markets where they are exposed to carbon price and regulatory risks for the first time. They may participate voluntarily to position their services or products as sustainable or carbon neutral as a differentiator. They will also need to calculate and manage carbon footprints likely reporting this information to a regulator at some point. Neither is it just those in the wholesale energy side that are impacted. The opportunities afforded and challenges to be managed also impact distribution where we can expect many changes as well around smart grids, smart devices, small-scale distributed renewables, EVs and much more. Here, we are likely to see aggregation services like the virtual power plant initiatives in the wholesale sector.

The energy transition represents an ongoing paradigm shift in the industry that is being rapidly pushed by politicians and regulatory bodies. Creating new risks, challenges, and opportunities, as we have already seen, other events can add further complexity and unpredictability. It also has an impact on the software solutions used by impacted businesses, particularly CTRM and related software.

EVOLVING REQUIREMENTS

A host of new requirements have already occurred and many more will emerge during the energy transition, from more frequent and automated trading and dispatch, through carbon footprint monitoring, management, and reporting through to the management and optimization of renewable generation facilities. Many of these involve CTRM and related software solutions.

These will also encompass the entire gamut of requirements from the need to manage new trading instruments, automate and optimize business processes, connect to new markets and more. However, there are three areas that this paper will focus upon as areas of opportunity and challenge as a part of this energy transition,

1. Biofuels and renewable fuels certificates,
2. Renewable power and green power certificates, and
3. Carbon markets (voluntary and mandatory).

These three areas have different renewable energy strategies resulting in distinguished needs around, managing risk, and optimizing operations in a low-carbon future. Each of these areas also requires dedicated, incremental functionality within the CTRM solution used to help manage the business through these challenges to unlock the opportunities and requires attention be paid to evolving business processes.

ION PREPARES FOR THE CHALLENGE

ION's position in the energy industry with more than 1,200 customers and over 30,000 users means that it has good visibility and insight into the challenges faced by energy and energy-related businesses.

With a set of products ranging from its cloud-enabled Aspect and Carbon zero through Allegro, Openlink, RightAngle and TriplePoint (amongst others), it is busy working with its customers to aid in managing these opportunities and challenges associated with adapting to the demands of the energy transition. This affords it diverse views on these problems and solutions and

allows ION to build out expertise and best practice around its software products.

To those ends, ION has been working with its customers to achieve several goals. One of these has been to ensure that its established power and hydrocarbon coverage has been combined with up-to-

date renewables capabilities, so that customers can diversify traditional energy portfolios with renewables in a single solution. It has also dedicated knowhow and capability in ensuring that it can deal with the renewables market needs by providing dedicated capabilities for renewables and environmental credits. Efficiency and agility are both key attributes of CTRM and related software in these markets as firms respond to shifting wants and needs, here ION has worked on ensuring it can automate high-volume, lower margin trading activities, and adapt to markets demands with flexible customer-centric solutions. A further area of investment has been in data management and transparency so that it can handle large granular data sets as well as provide sophisticated analytics to control business as well as enable decision support.

ION has focused on the three areas above in collaboration with customers helping them to overcome the challenges and realize benefit from the opportunities presented. In the areas of biofuels and renewable fuels certificates, it is working on challenges such as attached and detached trading (certificate trading combined with fuel (attached) or independent certificate trading (detached)). Other challenges included the need to track certificate IDs, manage the trade lifecycles, reconciliation across trading activities, avoidance of operational issues, fake certificates, and associated PR issues. There is also a need to use production forecasts to drive fuel and certificate position planning, reporting and decision making.

In this area too, ION can re-use their solutions strong supply chain capabilities to monitor physical activities

including support of different modes of transport, blending activity and inventory management. It has been able to combine the needs for those with both “traditional” and renewable operations into a single solution that manages both hydrocarbons and renewable fuels while providing the ability to optimize along the supply chain with a detailed view on forward delivery as well as inventory of fuel and certificates.

Another area is in renewable power and green power certificates. Here, ION has also ensured that it can provide a single solution for both renewables and fossil fuels-based power business that allows for proper contract operations and portfolio risk management. In terms of certificates, it provides a system of record for certificate lifecycle management. This ensures that PPAs can be properly managed including resulting power and certificates positions. For PPAs in particular, the handling of complex and non-linear payout structures, regional variations, and volume management and, enabling the operational management of such complex contracts as well as the ability to risk management the portfolio with typical analytics including Earnings at Risk / Profit at risk catering to senior management and investors reporting requirements..

Finally, in carbon markets, it has worked with customers to leverage its software to develop carbon strategies helping customers avoid and reduce carbon emissions, manage ESG needs with control over renewables investment via integrated and/or stand-alone software solutions within its portfolio. This has aided customers in analyzing and measuring their carbon footprint,

ensure transparency around its certificate and carbon avoidance activities and to avoid “green washing” and the reputational risk that involves, while providing the risk analytics required to manage exposures.

ION continues to work with its customers to ensure that

its solutions are the chosen solutions to help manage the opportunities and challenges of the ongoing energy transition. It has solutions to help all tiers and segments of the industry meet the challenges posed by the energy transition including those on the periphery who need to manage their carbon footprint.

SUMMARY

The energy transition is adding complexity, challenge and opportunity within the energy industry that require deployed software solutions like CTRM and related software to be agile and flexible. New requirements emerge, sometimes suddenly, that need to cater for, and business processes need to be continually optimized placing stress on the underlying solutions

deployed. ION is working with its wide range of customers – large and small – to ensure that its solutions meet the evolving needs of the industry now, and in the years to come. ION has the right amount of solution flexibility to adapt to changing market requirements and to enable its client base to take advantage of and manage emerging “Green Business” opportunities.

ABOUT **ION** COMMODITIES

We're transforming the world of commodities through innovation. Our mission is to understand the diverse and changing needs of your energy and commodities business and meet those needs with advanced technology. Wherever you are, and whatever industry you're in, ION Commodities solutions put you in complete control. With decades of knowledge and experience, we can provide proven solutions to the challenges you're facing. That's why over 30,000 users worldwide have partnered with ION to sharpen their decision-making and boost their productivity.



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ABOUT

Commodity Technology Advisory LLC

Commodity Technology Advisory is the leading analyst organization covering the ETRM and CTRM markets. We provide the invaluable insights into the issues and trends affecting the users and providers of the technologies that are crucial for success in the constantly evolving global commodities markets.

Patrick Reames and Gary Vasey head our team, whose combined 60-plus years in the energy and commodities markets, provides depth of understanding of the market and its issues that is unmatched and unrivaled by any analyst group.

For more information, please visit:

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